How are mass extinctions shaping biodiversity?

Confirmation Review

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Part I

Abstract

Part II

Literature Review

1 Mass extinctions

Extinction is a common process throughout Earth history; 99% of species that have ever existed are now extinct (Raup, 1981). However, extinction rates are not constant and occur in pulses across background taxonomic turnover (Raup and Sepkoski, 1982). Extinction events that wipe out over 75% of species are termed *mass extinctions* and have occurred five times over the last 500 million years (Raup and Sepkoski, 1982). On a macroevolutionary scale, mass extinctions mark the most dramatic species composition transitions in Earth history (Jablonski, 2001). Entire branches are trimmed from the tree of life, while the evolutionary trajectories of survivors are altered in the wake of a transformed global biota (Hull, 2015).

1.1 Rebound dynamics

The aftermath of a mass extinction has typically been described as 'recovery', yet Jablonski and Edie (2025) describe this as a misnomer, since this implies a return to the prior pre-extinction state. Instead, biodiversity rebounds are multifaceted and the response from the survivors of a mass extinction event may take several forms, transforming the post-extinction biota (Hull, 2015). For example, trends existing pre-mass extinction may continue unperturbed, including escalations (Hansen et al., 1999) and declines (Raup, 1981). Alternatively, the aftermath of a mass extinction can trigger diversification as surviving clades are offered ecological opportunity, filling the vacant niches left by extinct lineages.

Rebound dynamics can differ by the metric of diversity considered. A discordance between taxonomic and morphological diversity, which can be considered a proxy of ecological diversity, is common in empirical macroevolutionary studies. Intuitively, a diversifying clade would exhibit simultaneous taxonomic and phenotypic diversification, which are expected to follow a pattern of early high rates that decrease through time as niches become saturated (Schluter, 2000). However, empirical evidence often finds that peak taxonomic and morphological diversity are reached at different points in a clade's history (Foote, 1993; Hopkins, 2013; Ruta et al., 2013; Wan et al., 2021). This discordance has been observed across mass extinction events, such as in graptolites across the end-Ordovician mass extinction (Bapst et al., 2012), or therapsids across the Permian-Triassic mass extinction (Ruta et al., 2013). It has been suggested that functional diversity may be density-independent across mass extinction events (Edie et al., 2018). Therefore, understanding the rebound dynamics of clades based solely on taxic diversity metrics may not encapsulate the full complexity of rebound dynamics (Puttick et al., 2020).

2 Morphology, Disparity and Extinction

Disparity, the term used to describe morphological variation, has been widely used to investigate patterns of evolution in extant and extinct clades (Wills et al., 1994). Disparity can be defined as a way of describing the 'morphospace' (a multidimensional space configured by morphological similarity) that taxonomic units occupy (Guillerme et al., 2020). Changes in morphospace through time are often closely tied to shifts in ecological diversity. For example, expansion in morphospace, measured by variance-based metrics, may indicate a clade undergoing an adaptive radiation. Alternatively, a clade may move position within the morphospace, reflecting directional trends through time.

2.1 Extinction selectivity

Mass extinctions pose numerous scenarios of disparity (Figure 1). In adaptive scenarios, disparity may increase in opportunistic clades that diversify during the period of ecological opportunity following a mass extinction. Random extinction will reduce the total number of taxa, but have little effect on variance-based disparity metrics. Selective extinction, whereby taxa are more susceptible to extinction due to particular trait combinations, will have a detectable effect on variance-based measures (Foote, 1997; Korn et al., 2013). Korn et al. (2013) proposed several patterns to describe different selection processes: lateral (asymmetric selection that trims one side of the morphospace), marginal (symmetric selection that removes peripheral taxa) and non-selective (random process independent of the variables defining the morphospace). Work on contemporary extinctions has also investigated extinction selectivity, with certain trait combinations making extant taxa more susceptible to extinction (Ali et al., 2023; Matthews et al., 2024).

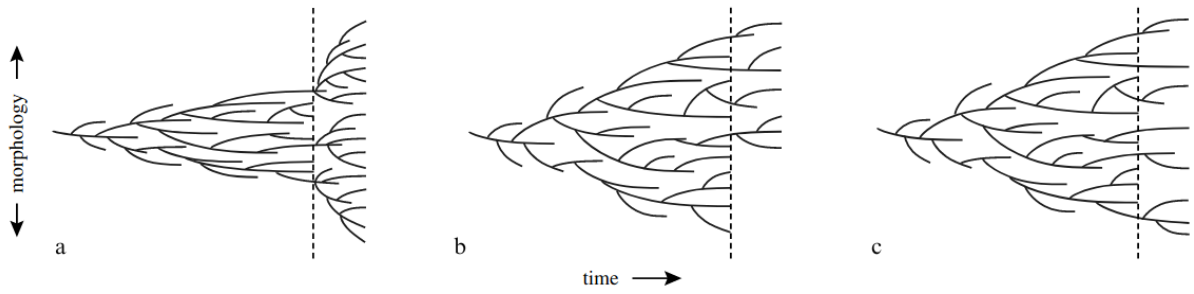

Figure 1: Three scenarios that may occur within a clade across a mass extinction (denoted by dashed line). a) adaptive radiation. b) selective extinction. c) non-selective extinction. Adapted from Grossnickle and Newham (2016).

2.2 Eutherian disparity across K-Pg Mass Extinction

The rebound dynamics of eutherian mammals (placentals and extinct relatives) across the K-Pg mass extinction is one of the most well-studied yet contentious inquiries of macroevolutionary biology. Eutherian mammals prior to the K-Pg boundary have been traditionally considered depauperate, small sized and limited in ecological variation, which contrasts the diversity seen in extant placentals. The widely held view until recently, known as the suppression hypothesis, follows that the extinction of non-avian dinosaurs released eutherian mammals from constraints of predation and competition, stimulating a radiation at the onset of the Cenozoic (Osborn, 1902; Alroy, 1999; Codron et al., 2013; O'Leary et al., 2013). The most significant morphological evidence for the suppression hypothesis in the fossil record concerns a shift in body size disparity (Alroy, 1999). Mesozoic mammals generally weighed less than 100 g and rarely over 5 kg (Benevento et al., 2023). The maximum body mass of mammals increased exponentially and by roughly three orders of magnitude by the late Eocene, which is consistent across continents and taxonomic groups (Cooper and Purvis, 2010; Smith et al., 2010; Raia et al., 2013; Slater, 2013).

While the suppression hypothesis is strongly supported by studies that have investigated changes in body size (Alroy, 1999; Slater, 2013), body size is limited as a metric for ecological variation since it can be influenced by several factors that may reflect stochastic rather than adaptive changes (Grossnickle, 2020). Recent studies instead focusing on morpho-functional traits (intrinsically linked to ecological function), such as jaw and dentition morphology, have instead suggested that eutherian diversity was attained through successive waves of diversification, starting far earlier than the K-Pg mass extinction (Grossnickle et al., 2019). The early rise hypothesis suggests an ecological diversification that occurred concurrently with the diversification of angiosperms and insects during and after the Cretaceous Terrestrial Revolution (125–80 Ma) (Figure 2) (Meredith et al., 2011; Benton et al., 2022). Ecological variation within Mesozoic mammals is not limited to diet; diversity in locomotor styles was reached as far back as the Jurassic in early mammaliaform groups, including early eutherians (Chen et al., 2019). Other studies have pointed towards the Paleocene-Eocene

Thermal Maximum (PETM), an episode of elevated global temperatures 56 Ma that widened the latitudinal range of angiosperm-dominated forest, which in turn spurred the dispersal and radiation of eutherians (Figure 2) (Bininda-Emonds et al., 2007). Benevento et al. (2019) found that eutherian mammals had the most significant expansion in jaw morphospace at the onset of the Eocene. While these alternative hypotheses do not necessarily refute the influence of the K-Pg mass extinction, they instead suggest the extinction accentuated evolutionary trajectories already underway.

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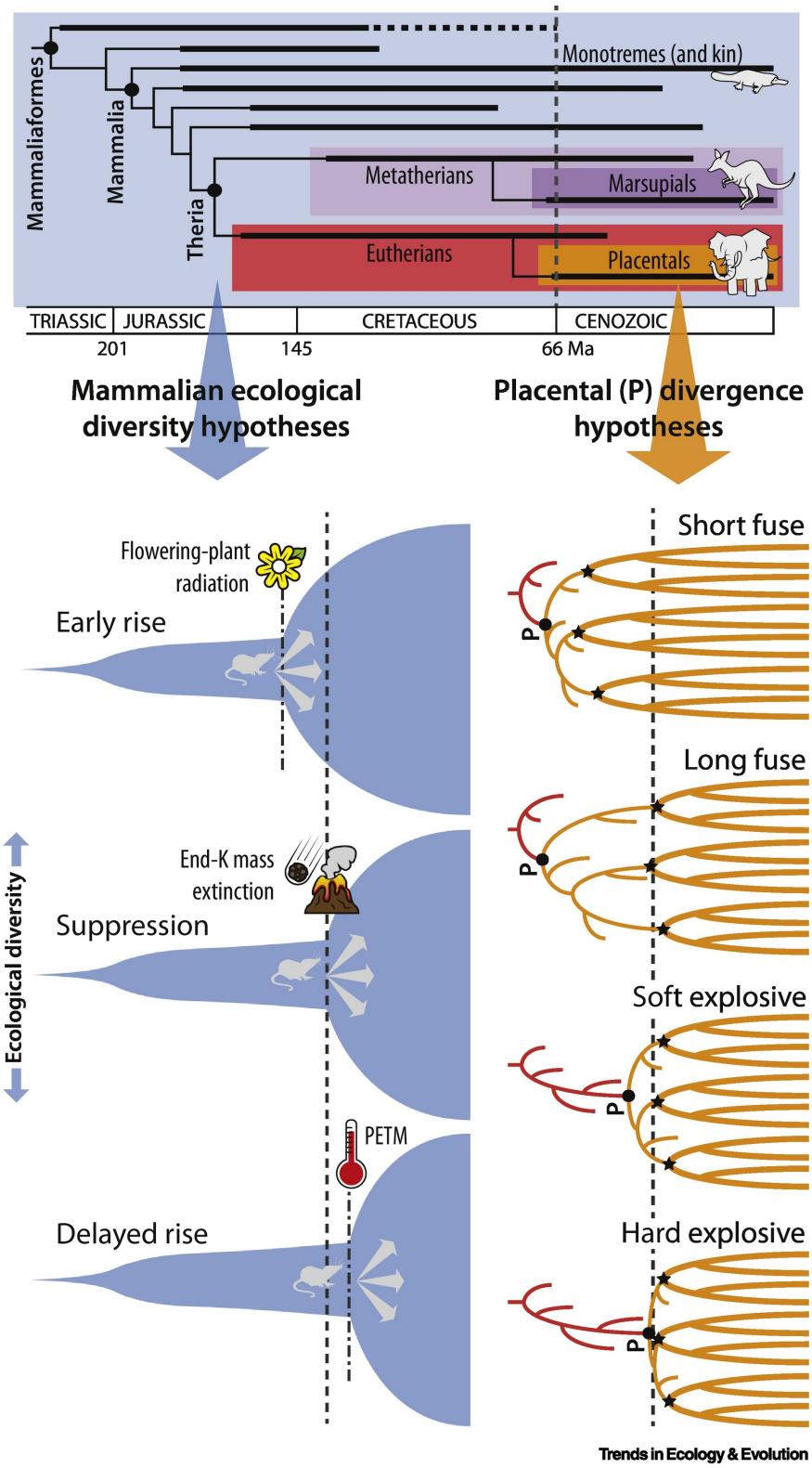


Figure 2: The contrasting ecological diversification triggers and placental origin hypotheses. Adapted from Grossnickle et al. (2019).

2.3 Disparity Analysis Methods

The conflicting findings of eutherian disparity across the K-Pg boundary raise the question: why do disparity analyses aiming to answer the same question arrive at different conclusions? Addressing this requires a critical look at the methodological underpinnings of disparity analyses.

For the purposes of this review, I focus on the most influential methodological axes that shape the outcomes of disparity analyses: (1) trait data, (2) disparity metric, (3) ancestral state estimation.

2.3.1 Types of trait data

The morphological data used in disparity analyses may fall into one of the following categories:

- **Discrete characters**, i.e. coding presence/absence or characteristic of traits. Discrete character matrices allow the inclusion of morphological features that are only present in a subset of the taxa. These datasets are often originally collected for phylogenetics and repurposed for disparity.
- **Continuous characters**, i.e. lengths of limbs, body mass, etc. More suited to investigations on a specific trait, rather than whole organism disparity.
- **Geometric morphometric**, i.e. homologous landmark data (e.g. Procrustes coordinates). More common for analyses at lower taxonomic levels, since need for homologous features in all specimens.

Given the varying scope of morphological variation that these contrasting approaches characterise, it is expected that they capture different aspects of disparity. For example, molar geometric morphometric data will only reveal variation related to diet/feeding ecology, whereas discrete character matrices of cranial and postcranial features may reveal variation in whole-organism morphology. Correlation between the outputs of these two approaches cannot be assumed, highlighting the discrepancies between disparity analyses. For example, Halliday and Goswami (2016), using a cladistic data matrix of dental, cranial and postcranial characters, found contrasting patterns of disparity across K-Pg compared to studies that used geometric morphometric analysis of molars (Wilson, 2013; Grossnickle and Newham, 2016). Furthermore, the overwhelming phylogenetic signal in cladistic discrete datasets, i.e. Halliday and Goswami (2016), may bury important functional variation contained within (Anderson and Friedman, 2012). Characters used during phylogenetics are selected to establish branching orders and differentiate taxa into distinct clades, which may artificially inflate disparity at nodes of cladogenesis (Gerber, 2019).

2.3.2 Disparity metrics

The contrasting findings may also be explained by different disparity metrics used between studies. For example, Halliday and Goswami (2016) found contrasting patterns of eutherian disparity between outputs using the sum of ranges compared to mean pairwise distance and sum of variances. Puttick et al. (2020) highlighted the importance of employing several different disparity metrics to explore different ways in which

the morphospace changes across a mass extinction. However, several studies investigating eutherian disparity have opted to use only a single disparity metric, often variance-based metrics such as sum of variances/ranges (Grossnickle and Polly, 2013; Wilson, 2013; Grossnickle and Newham, 2016). Puttick et al. (2020) highlighted the importance of employing several disparity metrics to explore different ways in which the morphospace changes across a mass extinction. T

2.3.3 Pipeline

Show the pipeline figure

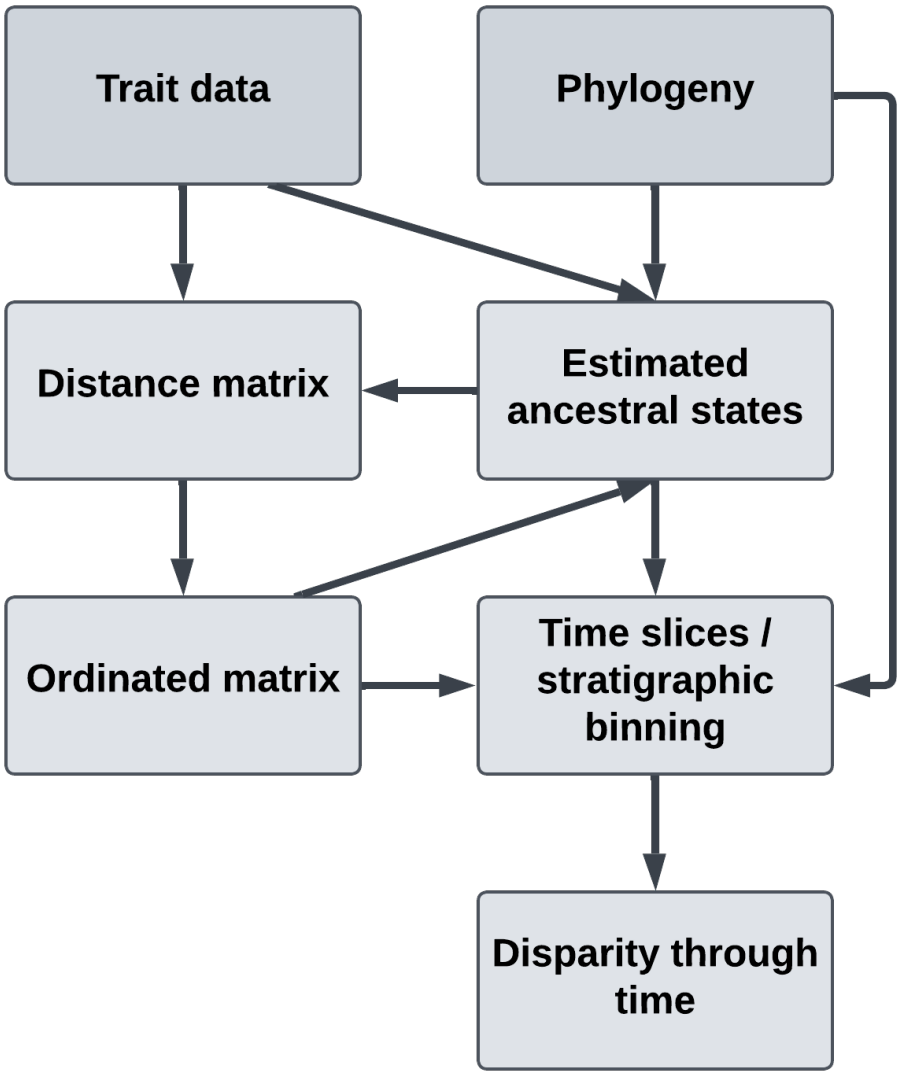


Figure 3: Disparity pipeline (N.B. this is specifically for discrete characters. Continuous characters would bypass the distance matrix after ASE or without ASE). As shown there are several routes to disparity, including bypassing ancestral state estimation (ASE), pre-ordination ASE, post-ordination ASE

3 Ancestral state estimation

The poor resolution of the fossil record is an inevitable limitation affecting disparity analyses, since only a small biased portion of a clade's evolutionary history is preserved. The incorporation of a phylogeny together with trait data enables a switch from simply observing a pattern (points in morphospace) to inferring a process through time, through the use of ancestral state estimation (ASE) (Lloyd, 2018).

3.1 Application of ASE

Ancestral state estimation (ASE) is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for operational taxa of a phylogenetic tree (Schluter et al., 1997; Cunningham et al., 1998; Yang, 2006; Revell, 2025). Trait data may be continuous, such as body size, or discrete, such as diet type or presence/absence of a character. The application of ASE has been widely utilised in phylogenetic comparative methods, including disparity.

Intuitively, ASE is appealing for those studying deep-time patterns of disparity. Incorporating a phylogeny into disparity analyses will unavoidably introduce unsampled 'ghost lineages' (Brusatte et al., 2011). A ghost lineage is the hypothetical ancestor that is not observed in the fossil record, but can be inferred to have existed due to the presence of other fossils/extant taxa. This can be best explained as the branch on a phylogeny between two observed taxonomic units (Figure 4).

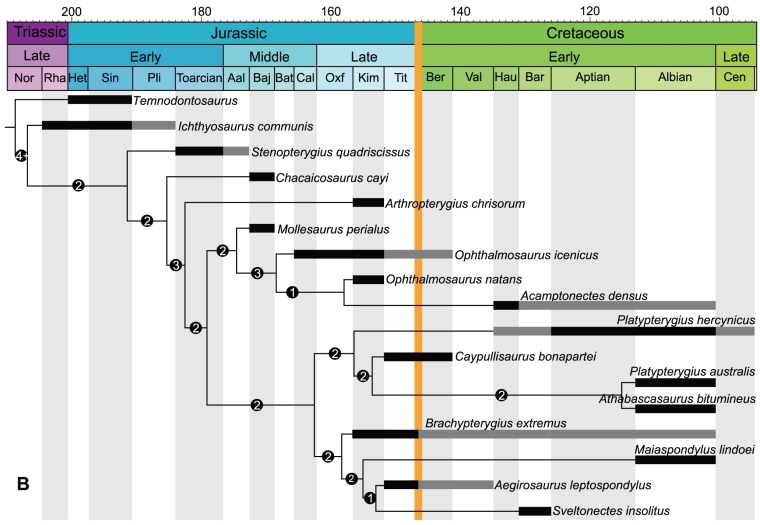


Figure 4: Phylogeny of ichthyosaurs showing observed stratigraphic range for each species (thick lines) and inferred ghost lineages (thin lines) (Fischer et al., 2012).

ASE has been employed as a way of inferring the traits of these hypothetical species based on the traits of their descendents. During ASE, the estimated traits for nodes of a phylogeny are used as a way of sampling these ghost lineages. This fills in the gaps in the fossil record and alleviates the bias associated with uneven sampling across the phylogenetic tree (Friedman, 2009; Brusatte et al., 2011).

3.1.1 Discrete characters

Discrete characters are typically analysed using parsimony or likelihood approaches, although likelihood methods are generally preferred. Likelihood approaches, such as Bayesian inference or Maximum likelihood, optimize solutions based on the tree topology, branch lengths and the model of character evolution. Discrete characters are modelled using a continuous time Markov chain process, known as the Mk model (where $k =$

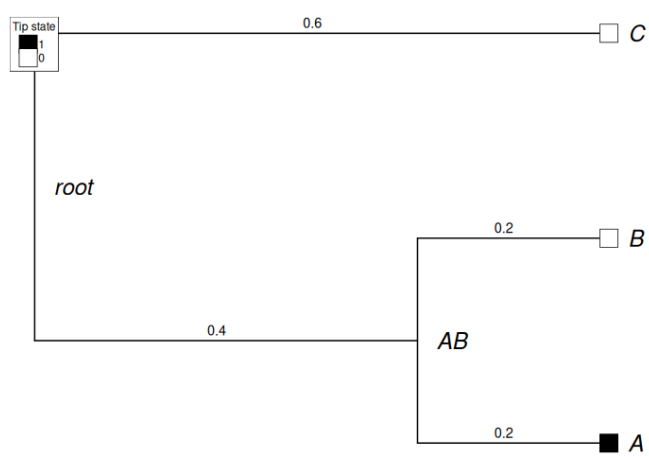
number of possible states) (Pagel, 1994; Yang, 2006). This process is specified by a $k \times k$ matrix, $\mathbf{Q}$, which gives the instantaneous transition rates (q) between states for each character.

$$\mathbf{Q} = \begin{bmatrix} -q_{0,1} & q_{0,1} \\ q_{1,0} & -q_{1,0} \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 \\ 0.5 & -0.5 \end{bmatrix}$$

There are a number of Mk transition rate models that can be fit to data. The 'equal-rates model' (ER) is the simplest and estimates just one parameter, since it assumes all transitions occur with equal probability, equivalent to the Jukes-Cantor model of molecular evolution. The assumptions of the ER model can be eased with the 'symmetric' model (SYM): rate of change between any two character states is equal in the forwards and backwards direction, but may differ between different pairs of states. The 'all-rates-different' (ARD) model allows for a different transition rate in each direction, includes a total of $k \times (k-1)$ parameters to estimate. The probability of each transition at any given interval of time (t) can be calculated by

$$\mathbf{P} = \exp(\mathbf{Q}t)$$

So if we want to calculate probability of each transition along the tree branch from root to node AB, $t = 0.4$.



$$\mathbf{P}_{t=0.4} = \exp(\mathbf{Q} \times 0.4) = \begin{bmatrix} 0.835 & 0.164 \\ 0.164 & 0.835 \end{bmatrix}$$

One further probability to consider is the 'root prior', which is essentially whether the root node of the tree was in state 0 or 1. This is typically set with a flat root prior $\pi_0 = \pi_1 = 0.5$.

The most popular method of ASE is *marginal* estimation, which involves proceeding from node to node on the phylogeny, and, at each node, computing the probability of observing the tip data of our tree conditioned

on fixing the node we've visited to each one of the set of distinct values of our trait. If we were to compute the probability that the root was in state 0,

For example, if we were to calculate the probability of the root node was in state 0:

$$\begin{aligned}
P(AB = 0) &= \pi_0 \times P(0|0, t_{AB}) \times P(1|0, t_A) \times P(0|0, t_B) \times P(0|0, t_C) \\
&\quad + \pi_1 \times P(0|1, t_{AB}) \times P(1|0, t_A) \times P(0|0, t_B) \times P(0|1, t_C) \\
&= 0.5 \times 0.0532 + 0.5 \times 0.0034 \\
&= 0.0266 + 0.0017 = 0.0283
\end{aligned}$$

$$\begin{aligned}
P(AB = 1) &= \pi_0 \times P(1|0, t_{AB}) \times P(1|1, t_A) \times P(0|1, t_B) \times P(0|0, t_C) \\
&\quad + \pi_1 \times P(1|1, t_{AB}) \times P(1|1, t_A) \times P(0|1, t_B) \times P(0|1, t_C) \\
&= 0.5 \times 0.0105 + 0.5 \times 0.0159 \\
&= 0.0053 + 0.0080 = 0.0133
\end{aligned}$$

These marginal likelihoods are then scaled by their sum, giving $P(0,\ 1) = [0.68,\ 0.32]$. In marginal estimation, this is calculated node by node. Since this can become extremely complicated as the phylogeny increases in size, in practice functions use algorithms such as Felsenstein's pruning algorithm (Felsenstein, 1981).

3.1.2 Continuous characters

Continuous trait evolution is typically modelled under a Brownian motion (BM) model of evolution (Felsenstein, 1985; O'Meara et al., 2006). Brownian motion models trait evolution as a directionless, stochastic process under which continuous characters undergo changes along the branches of a phylogeny. These evolutionary changes come from a Gaussian distribution with a mean of 0 and variance proportional to the product of the rate of evolution and time elapsed ($\sigma^2 t$). Therefore, the expected amount of trait divergence between any two taxa is proportional to the amount of shared evolutionary history according to the phylogeny.

Under BM, the vector ($\mathbf{x}$) of observed tip trait values follows a multivariate normal (MVN) distribution and has an expected value that is equal to the root state x_0.

$$\boldsymbol{x} \sim \mathrm{MVN}(x_0,\ \sigma^2 \boldsymbol{C})$$

$\boldsymbol{C}$ is a variance-covariance matrix of N dimensions where the diagonal elements $\boldsymbol{C}_{ii}$ are the tree heights from root to tip i and $\boldsymbol{C}_{ij}$ is the height above the root of the most recent common ancestor of i and j. When estimating ancestral states under BM, the trait values of the internal nodes are also distributed within this MVN distribution. Therefore, the $\boldsymbol{C}$ matrix can be expanded by one column and row for each node (except the root). During Maximum Likelihood, the x_0 and σ^2 parameters are estimated as the values that maximize the likelihood of observing the tip vector. Internal node values can be estimated using the conditional

expectations of this distribution, i.e. the trait value at an internal node is calculated as a weighted average of the observed tip values, determined by the amount of shared evolutionary history (given by $\boldsymbol{C}$) and the node's position in the tree.

3.2 Challenges of ASE

'Character state reconstructions can provide a powerful mechanism for studying many facets of the evolutionary process. However, the zeal with which these techniques are sometimes advocated belies the complexity of the problem' (Swofford and Maddison, 1992).

3.2.1 Uncertainty in estimates

A critical issue within ancestral state estimation is that estimates tend to have broad uncertainty (Schluter et al., 1997). This uncertainty is not uniform across the tree: it typically increases towards root of the tree, where direct information from observed tip data becomes increasingly diffuse (Gascuel and Steel, 2014). For discrete characters, this uncertainty manifests as inconclusive scaled likelihoods, i.e., state 0 = 0.51, state 1 = 0.49. For continuous characters, uncertainty is wide confidence intervals.

How this uncertainty in estimations should be dealt with is equivocal. Despite the encouragement for uncertainty to be carried forward in comparative analyses (Silvestro et al., 2015), most computational implementations of ASE return point estimates. For continuous characters, this will be the median value of the confidence interval and for discrete characters, it will be the most probable state according to a threshold (see later). By ignoring uncertainty, point estimates can lead to overconfidence in results and erroneous inferences in downstream analyses (such as down the disparity pipeline), all while obscuring potentially 'accurate' uncertainty.

For discrete characters, there is a level of subjectivity in how to collapse the states into the most probable based on their scaled likelihood. It is common to have a threshold implementation in ASE software, whereby the user can set a scaled likelihood at which point states are collapsed to a point estimate, or left as uncertain, which is denoted as '0/1', for example. However, these thresholds are often arbitrarily set with no guidance of what is the 'correct' value (Smith and Donoghue, 2022; Casali et al., 2023). Much of the subjectivity relates to the trade-off between permissiveness and conservativeness; higher thresholds are likely too conservative and so too many states are left uncertain, perhaps missing true signals. Whereas lower thresholds may be too permissive and can lead to erroneous inferences. Interestingly, Keating (2023) highlights that uncertainty returned by a specified model is a better predictor of accuracy than likelihood diagnostics, such as AIC.

3.2.2 Sensitivity to fossil sampling

Given that uncertainty is often high in ASE, how then can estimates be improved? Ané (2008) demonstrates that ASE is *statistically inconsistent*, meaning that even if the sample size of extant tips is infinite, estimates are not expected to converge on the true value. Instead, the inclusion of fossils can improve estimates, due

to the shorter evolutionary path to the root of the tree (Finarelli and Flynn, 2006; Finarelli and Goswami, 2013). Fossils can be used as priors on nodes, or simply added to the tree as tips.

By extension, inclusion of fossils can greatly influence inferences into disparity. Extant-only datasets often miss early rapid morphological diversification signals (Harmon et al., 2010), whereas analyses incorporating fossil data often finds the reverse (Hughes et al., 2013) For example, Mitchell (2015) found that analyses of bird disparity were underestimated when using ancestral state estimation of extant-only datasets.

3.2.3 Model dependency

Another key challenge with ASE is its sensitivity to violations of its model assumptions. For continuous characters, this is particularly pertinent when traits have evolved under non-BM (Revell, 2025). When trait evolution deviates from BM—due to directional trends, bounded evolution, or regime shifts—ASE can return biased estimations of ancestral states.

Both Webster and Purvis (2002) and Finarelli and Flynn (2006) were unable to recover accurate ancestral estimates of body size in planktonic foraminifera and mammals, respectively, due to a trend of increase through time. This violates the assumption of zero-mean change under BM. Similar issues arise when the trait data evolved under an Ornstein Uhlenbeck model, whereby trait data evolves in a Brownian way but under a stabilizing selection (Royer-Carenzi and Didier, 2016).

It is common for regime shifts to occur at mass extinction boundaries (Jablonski, 1986). For example, Slater (2013) found that mammal body size was relatively constrained (OU) up until the K-Pg mass extinction which then allowed a release of constraint and Brownian motion evolution.

As discussed previously, fossil sampling is crucial for accurate ASE, especially when model assumptions are violated (Puttick, 2016). Fossils assist in estimating and constraining model parameters, as well as distinguishing different evolutionary processes (Goswami and Clavel, 2025). For example, when estimating states under an OU model, fossils are necessary for estimating the primary optimum. Similarly, fossils would be necessary to detect a trend, such as Cope's rule. A similar logic applies to discrete characters.

Keating (2023) found that inclusion of fossils improved ASE estimates, irrespective of model misspecification.

3.2.4 Pre- or post-ordination

ASE can be performed at two points in the disparity analysis pipeline (Figure 3): either pre-ordination or post-ordination. Pre-ordination ASE involves estimating the ancestral states of the raw trait data, whereas post-ordination ASE estimates the PC scores of ancestral nodes. The pre-ordination has become favoured (Lloyd, 2018) due to various statistical issues with post-ordination ASE. For example, post-ordination ASE means ancestral values have to fall within the convex hull of the observed trait data (Stayton, 2015). While this is usually biologically realistic due to the zero-force evolutionary law (McShea and Brandon, 2010), in a scenario where taxon sampling is poor, true ancestral states may fall outside the morphological extremes des-

ignated by observed data. Lloyd (2018) found that the post-ordination ASE method recovers an implausibly low amount of convergent evolution and patterns dictated more so by phylogenetic signal than morphological signal. Despite this, there are some practical benefits to performing post-ordination ASE. Firstly, it is less computationally expensive. Secondly, it mitigates issues with trait correlations. ASE methods assume trait independence, which is rare in morphological trait datasets (e.g., body size and limb length are positively correlated). Pre-ordination ASE, by ignoring this correlation, may return biologically implausible estimations that violate allometry, for example. Through ordination, the dominant sources of trait correlation are captured within the first few PC axes. The orthogonality of PC axes means zero co-variance between 'traits' (which are now PC axes) thus reducing the violation of the assumption of independence.

4 Detecting shifts in disparity

5 Aims and objectives

Knowledge gap

Part III

Thesis plan

6 Chapter One: What is the effect of ancestral state estimations on disparity?

6.1 Introduction

- Morphology can tell us interesting patterns about evolutionary processes.
- The combination of morphological datasets (using fossils) and phylogenies means we can infer these processes through deep time
- This is particularly useful when studying major evolutionary events (such as a mass extinction)
- However, the fossil record is incomplete and this restricts our ability to look at deep time patterns of morphology.
- Many studies have made use of ancestral state estimation (ASE) to fill in gaps left by 'ghost lineages'. We know some species must have existed between root and tips but we don't know what they looked like.

- Particularly has been implemented in studies looking at disparity, which describes the multidimensional 'morphospace' in which taxa occupy.
- ASE can be used to estimate the character state values (both continuous and discrete characters) at nodes on a phylogeny, using observed trait data (from living and fossil specimens).

6.1.1 Why does this matter?

- ASE estimates are often shit and a 'glorified average'. But, what can we do with them?
- Erroneous estimates can mislead values of disparity and therefore macroevolutionary inferences.

6.2 Methods

6.2.1 Simulations

- I simulated a set of 100 trees using `treats` (Guillerme, 2024). These trees were simulated with birth-death parameters as $\lambda = 1.0$, and $\mu = 0.7$. This leaves trees with a range of 50-200 fossil tips and 50 extant tips.
- I simulated discrete and continuous trait data using `map.traits`. This effectively maps the 'true' evolutionary history across the phylogeny (tips and nodes have states mapped).
- For discrete traits, I simulated trait data across each tree (100 characters, 85% binary, 15% 3-state) using Mk model with 3 different transition rates: fast = 0.9, medium = 0.1, slow = 0.01.
- For continuous traits, I simulated BM and OU model trait data (50 characters).

6.2.2 Fossil sampling

- I then removed all node states from the matrices, leaving just tip states.
- In order to understand how ASE accuracy is affected by fossil sampling level, I pruned the fossil tips at 5 levels, preserving the root to maintain tree height across each level:
- **Living**: only extant tips are retained
- **Low**: 5% fossils retained
- **Medium**: 15% fossils retained
- **High**: 50% fossils retained
- **All**: All fossils retained

6.2.3 ASE

- I then ran ASE on these matrices using `multi.ace`, with three different parameters (for discrete states).
- **Relative majority rule**: the ancestral state estimated are all the ones that are have a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states:

$$\text{estimated state} \geq \left(\max(\text{likelihood}) - \frac{1}{n_{\text{states}}}\right)$$

  If more than one state is over this threshold, it is recovered as uncertain e.g., (0/1).
- **Strict majority rule**: Setting threshold parameter to a set value will select the ancestral states estimate with a scaled likelihood equal or greater than this designated value (in this case I did 0.5).
- **No majority rule (sampling)**: `multi.ace` has a `sample = x` parameter, whereby you can return x matrices, in which states returned are by the scaled likelihood of each state appearing at each node. In this case I used sample = 10 for fast computation (in future I will do 100/1000).
- Ran `multi.ace` on continuous characters using BM as model.
- I will also do post-ordination ace to see how that compares to pre-ordination ace.

6.2.4 Raw distance comparison between true vs ace

- I then used a Hamming distance metric to compare corresponding nodes between the 'true' vs 'aced' matrices.
- Distances were calculated only for nodes that were present in all fossil sampling levels to allow for comparison.
- The distance assigned a penalty of 1 for any wrong estimated state and also for uncertainty (i.e 0/1).
- For continuous traits, I firstly scaled each matrix and then measured the euclidean distance between true vs ace for each corresponding node.
- I also looked at the distance at just the root, to see how well ace recovers true value at deepest node.

6.2.5 Distance comparison between ordinated true vs ace

- I then ordinated the true matrix (distance matrix then PCO for discrete, PCA for continuous), then projected the aced matrices onto this true matrix.
- I measured the Euclidean distance between corresponding nodes again.

6.2.6 Difference in disparity between true vs ace

- from the ordinated matrices, I calculated (will choose which metrics, at the moment sum variance and ranges and average nearest neighbour distance) disparity and calculated the difference between true vs aced.

6.3 Results (or what I expect to find)

6.3.1 Raw distances

- Slower transition rates = ASE accuracy improves
- Increasing fossil sampling improves ASE accuracy (for discrete traits), particularly at deeper nodes.
- Using sample method of ASE improved estimate accuracy
- Using sample method is relatively insensitive to low fossil sampling and finds high accuracy at all fossil sampling levels
- At the root node, the impact of fossil sampling becomes more pertinent i.e., higher fossil sampling improves estimates
- For continuous traits, ASE accuracy was much better for BM traits than OU
- Estimates of continuous traits stay relatively constant across fossil preservation levels.
- For continuous traits, using point estimates was more accurate than sample method.

6.3.2 Ordination

- I expect that patterns of distance to true value will correspond to the raw distance pattern (more fossil sampling = higher ASE accuracy, sample method = higher ASE accuracy)
- I expect that fast transition rates will exhibit clustering at the centre of the morphospace, particularly with the default multi.ace parameter, due to the high amount of uncertainty recovered at deeper nodes. I think that the sample method of ASE will alleviate this artefact by removing this uncertainty.

6.3.3 Disparity

- Preliminary results have found that using sample ASE method recovers high accuracy for all transition rates.
- Using the default ASE method recovers much lower accuracy, particularly with low fossil sampling.

- I expect to see different results for different disparity metrics. For example, metrics that focus on density will be able to identify these artifacts of clustering due to uncertainty in node estimates, which variance based metrics will miss.
- This will highlight the need for multiple different disparity metrics to be used in disparity analyses

7 Chapter Two: Measuring changes in trends in disparity through time

8 Chapter Three: Eutherian disparity over K-Pg mass extinction

Part IV

Summary of research achievements

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